This section includes the following topics:

- Define Lebesgue measurable functions
- Measurability of functions that are obtained by sums, products and compositions of some other measurable functions
- Function convergence
- Characterization of measurable functions: Simple Function Approximation
- Littlewood's Three Principles:
    - Every measurable set is nearly the union of a finite collection of open intervals
    - Every point-wise convergent sequence of functions is nearly uniform convergent
    - Every measurable function is nearly continuous

Define Lebesgue Measurable Functions

Proposition 1.9.1: E is a measurable set. f is an extended real-valued function (i.e. it takes values in $[-\infty, \infty]$) defined on E. The following statements are equivalent.

- (a) For any real number c, the set $\{x \in E \mid f(x) > c\}$ is measurable;
- (b) For any real number c, the set $\{x \in E \mid f(x) \geq c\}$ is measurable;
- (c) For any real number c, the set $\{x \in E \mid f(x) < c\}$ is measurable;
- (d) For any real number c, the set $\{x \in E \mid f(x) \leq c\}$ is measurable.

If any of these statement is true for f, we say that f is Lebesgue measurable.

Proof:

For $(a) \Rightarrow (b)$: Let $E_n = \{x \in E \mid f(x) > c - \frac{1}{n}\}$, for $n \geq 1$. Let $E_0 = \{x \in E \mid f(x) \geq c\}$. Then $E_0 = \bigcap_{n=1}^{\infty} E_n$. Since (a) is true for f, E_n is measurable for $n \geq 1$. Therefore, E_0 is measurable.

For $(b) \Rightarrow (c)$: The set $\{x \in E \mid f(x) < c\} = R - E_0$, and thus is measurable.

For $(c) \Rightarrow (d)$: This can be shown by a similar argument as in showing $(a) \Rightarrow (b)$.

For $(d) \Rightarrow (a)$: The set $\{x \in E \mid f(x) > c\} = R - \{x \in E \mid f(x) \leq c\}$ and thus is measurable.

Note that the above arguments hold even when c is $-\infty$ or ∞.

QED

Measurability of Functions

Proposition 1.9.2: A function f, defined on a measurable domain E, is measurable if and only if for any open set O, the pre-image $f^{-1}(O)$ is measurable.

Proof: If f is measurable, by Proposition 1.9.1, we can easily see that for every open interval I, the pre-image $f^{-1}(I)$ is measurable. Consider an open set O. It is a union of a countable collection of open intervals, i.e. $O = \bigcup_{n=1}^{\infty} I_n$. It is not difficult to see that

$$f^{-1}(O) = \bigcup_{n=1}^{\infty} f^{-1}(I_n)$$

, and thus is measurable.

Conversely, suppose $f^{-1}(O)$ is measurable for any open set O. Then for any real number c, $(-\infty, c)$ is an open set. Thus $f^{-1}((-\infty, c))$ is measurable and f is measurable.

QED

Proposition 1.9.3: A continuous function f, defined on a measurable domain E, is measurable.

Proof: Since f is continuous, $f^{-1}(O) = E \cap U$, where U is open for any open set O. Since U and E are measurable, so is $f^{-1}(O)$. By Proposition 1.9.2, f is measurable.

QED

Proposition 1.9.4: A monotone function f, defined on an interval, is measurable.

Proof: We defer the proof once we establish the measurability of point-wise convergence.

QED

Proposition 1.9.5: Let f be a function defined on a measurable domain E. Then

- (a) If f is measurable on E and $g = f$ a.e. on E, then g is also measurable on E
- (b) D_1 is a measurable subset of E. Then f is measurable, if and only if, f is measurable on both D_1 and $E - D_1$

Proof: (a) Let $E_0 = \{x \in E \mid g(x) \neq f(x)\}$. Then $m(E_0) = 0$. Consider a real number c. The set $\{x \in E \mid g(x) < c\} = E_1 \cup E_2$, where

$$E_1 = \{x \in E - E_0 \mid g(x) < c\}$$
$$E_2 = \{x \in E_0 \mid g(x) < c\}$$

Since $E_2 \subseteq E_0$, we have $m(E_2) = 0$. Note that $g(x) = f(x)$ on E_1 and f is measurable. So $E_3 = \{x \in E \mid f(x) < c\}$ is measurable. Then $E_1 = E_3 \cap (E - E_0)$ is also measurable. As a result, $E_1 \cup E_2$ is measurable. Thus, g is measurable on E.

(b) Suppose f is measurable. Consider a real number c. Let $G = \{x \in E \mid f(x) < c\}$.The set $G_1 = \{x \in D_1 \mid f(x) < c\}$ equals to $G \cap D_1$, and thus is measurable. Similarly, the set $G_2 = \{x \in E - D_1 \mid f(x) < c\}$ equals to $G \cap (E - D_1)$, and thus is measurable. Therefore, f is measurable on both D_1 and $E - D_1$.

Conversely, suppose f is measurable on both D_1 and $E - D_1$. Using the notation above, the set $G = G_1 \cup G_2$. Since G_1, G_2 are measurable, G is measurable. As a result, f is a measurable function on E.

QED

Proposition 1.9.6: Let f, g be real-valued measurable functions on E.

- (a) Linearity: For any α, β, the function $\alpha f + \beta g$ is measurable on E
- (b) Product: The function $f \cdot g$ is measurable.

Proof: (a) αf is clearly measurable. For measurability of $f + g$, it suffices to show that the set $G = \{x \in E \mid f(x) + g(x) < c\}$ is measurable for any real number c.

To this end, note that the set of ration numbers Q is dense in R. So for any x that satisfies $f(x) < c - g(x)$, there is a rational number q such that $f(x) < q < c - g(x)$. So the set

$$E = \bigcup_{q \in Q} [\{x \in E \| \ g(x) < c - q\} \cap \{x \in E \| \ f(x) < q\}]$$

. Since Q is countable, E is the union of a countable collection of measurable sets. Thus E is measurable.

(b) Note that $f \cdot g = \frac{1}{2}[(f+g)^2 - f^2 - g^2]$. We have established the linearity property. So it suffices to show that f^2 is measurable.

To this end, consider any $c \geq 0$. Then the set

$$\{x \in E \mid f^2(x) > c\} = \{x \in E \mid f(x) > \sqrt{c}\} \cup \{x \in E \mid f(x) < -\sqrt{c}\}$$

, and thus is measurable.

When $c < 0$, the set $\{x \in E \mid f^2(x) > c\} = E$, and of course is measurable.

QED

Proposition 1.9.7: g is a measurable function on E and f is a continuous function on R. Then the composition $f \circ g$ is measurable on E.

Proof: Consider any open set O. By Proposition 1.9.2, it suffices to show $(f \circ g)^{-1}(O)$ is measurable. To this end, note f is continuous on R. Thus $f^{-1}(O)$ is open. Thus $g^{-1}(f^{-1}(O))$ is measurable.

QED

Proposition 1.9.8: For a finite family of measurable functions $\{f_k\}_{k=1}^n$ on a common domain E, the functions $\max\{f_1, \ldots, f_n\}$ and $\min\{f_1, \ldots, f_n\}$ are also measurable.

Proof: It suffices to show the "max" case, because $\min\{f_1, \ldots, f_n\} = -\max\{-f_1, \ldots, -f_n\}$.

Let $f = \max\{f_1, \ldots, f_n\}$. Then,

$$\{x \in E \mid f(x) > c\} = \bigcup_{k=1}^{n} \{x \in E \mid f_k(x) > c\}$$

, and thus is measurable.

QED

As a result of Proposition 1.9.8, if f is measurable, the following functions are all measurable:

$$\begin{aligned} f^+ &= \max\{f, 0\} \\ f^- &= \max\{-f, 0\} \\ |f| &= \max\{f(x), -f(x)\} \end{aligned}$$

Function Convergence and Simple Function Approximation

When we would like to prove a function f is measurable, a common technique is to construct a convergent sequence of measurable functions $f_n \to f$ and then show the measurability of f.

Here we need to define the different notions of function convergence.

Let $\{f_n\}_{n=1}^{\infty}$ be a countable sequence of measurable functions defined on a common domain E. Let f be a function that is also defined on E.

Point-wise Convergence

f_n converges to f point-wisely, written as $f_n \to f$ p.w., if for every $\epsilon > 0$, there is an associated $N(x)$ for every $x \in E$ such that when $n \geq N(x)$ we have $|f_n(x) - f(x)| < \epsilon$.

In other words, for every $x \in E$, $f_n(x) \to f(x)$.

Point-wise Convergence almost everywhere

f_n converges to f point-wisely almost everywhere, written as $f_n \to f$ a.e., if f_n converges to f point-wisely on a measurable subset $E_1 \subseteq E$, and the remaining part $E_0 = E - E_1$ has measure 0.

Uniform Convergence

f_n converges to f uniformly, written as $f_n \to f$ u.f., if for every $\epsilon > 0$, there is a uniform N such that for every $x \in E$, when $n \geq N$ we have $|f_n(x) - f(x)| < \epsilon$.

In other words, $f_n(x) \to f(x)$ "in the same speed" for every $x \in E$.

Proposition 1.9.9: If $f_n \to f$ a.e., and f_n are measurable on E, then f is measurable on E.

Proof: We may assume that $f_n \to f$ p.w. everywhere. Because if $f_n \to f$ a.e. on E, then $f_n \to f$ p.w. everywhere on $E_1 \subseteq E$. Thus $E_0 = E - E_1$ has measure 0. Therefore, f is measurable on E_0 since any subset of E_0 is measurable. So if f is measurable as well on E_1, by Proposition 1.9.5(b), then f is also measurable on E.

So we proceed to show the measurability of f under the assumption that $f_n \to f$ p.w. on E.

Consider a real number c. For $x \in E$, $f_n(x) \to f(x) < c$, if and only if, there exists some m, k such that

$$f_n(x) < c - \frac{1}{k}, \;\; n \geq m$$

Therefore, the following sets are equal,

$$\{x \in E \mid f(x) < c\} = \bigcup_{0 \leq m, k \leq \infty} [\bigcap_{n \geq m} \{x \in E \mid f_n(x) < c - \frac{1}{k}\}]$$

Since every f_n is measurable, the right-hand side is formed by countable union and intersection of measurable sets, and thus is measurable. As a result, the left-hand side is also measurable. Therefore, f is a measurable function.

QED

We say that a function f defined on a measurable domain E is a **simple** function, if it is measurable on E and it takes a finite number of values.

A function is said to be a characteristic function of a set E, if it is defined on R and satisfies: $C_E(x) = 1$ when $x \in E$, and $C_E(x) = 0$ when $x \notin E$.

Proposition 1.9.10 (Simple Approximation Lemma): f is a measurable function on E. If f is bounded on E, then for any $\epsilon > 0$, there are simple functions ϕ and ψ such that

$$\phi \leq f \leq \psi, \;\; and \;\; 0 \leq \psi - \phi < \epsilon$$

Proof: Since f is bounded, there is a $M > 0$ such that $-M < f(x) < M$ for every $x \in E$. Let $N = \lceil \frac{2M}{\epsilon} \rceil + 1$. Then let $\epsilon_0 = \frac{2M}{N}$. Clearly $\epsilon_0 < \epsilon$. Construct the sequence $a_n, \; n = 1, \ldots, N+1$, where $a_1 = -M$ and $a_{n+1} = a_n + \epsilon_0$.

Since f is measurable on E, for every interval $I_n = [a_n, a_{n+1})$, $E_n = f^{-1}(I_n)$ is measurable. Construct the simple functions $\phi = \sum_{n=1}^{N} a_n \cdot C_{E_n}$ and $\psi = \sum_{n=1}^{N} a_{n+1} \cdot C_{E_n}$ as desired.

QED

If f is unbounded, we can still construct a sequence of simple functions that converges to it point-wisely.

Proposition 1.9.11 (Simple Approximation Theorem): f is a real-valued function on E. Then f is measurable, if and only if, there is a sequence of simple functions $\{\phi_n\}$ such that $\phi_n \to f$ p.w., and $|\phi_n| \leq |f|$ for every n.

Proof: The direction from right to left is taken care of by Proposition 1.9.9.

For the other direction, assume f is measurable on E. We may assume f is positive. For every natural number n, let $E_n = \{x \in E \mid f(x) \leq n\}$. Then E_n is measurable. The function f restricted to E_n is also measurable and bounded. By the Simple Approximation Lemma, there are simple functions g_n, h_n on E_n such that $g_n \leq f \leq h_n$ and $0 \leq h_n - g_n < \frac{1}{n}$. Construct a simple function as follows:

$$\phi_n(x) = \begin{cases} n, & f(x) > n \\ g_n(x), & f(x) \leq n \end{cases}$$

Clearly, $|g_n| \leq |f|$ for every n. In addition, it is easy to see that $g_n \to f$ p.w.

Finally, consider the generate case where f can take both positive, negative and 0 values. Note that $f = f^+ - f^- + f_0$. Both f^+ and f^- are positive cases, and thus there exist simple functions ϕ_n^+ and ϕ_n^- such that converge to f^+ and f^- point-wisely, respectively. Furthermore, $|\phi_n^+| \leq |f^+|$ and $|\phi_n^-| \leq |f^-|$. For f_0, we construct the simple function $\phi' = 0$ on E. Finally, take the simple function

$$\phi_n = \phi_n^+ - \phi_n^- + \phi'$$

, and it is what we desire.

QED

Littlewood's Three Principles

The following Egoroff's Theorem states that every point-wise convergence is nearly uniform convergence

And the Lustin's Theorem states that every measurable function is nearly a continuous function.

Egoroff's Theorem

Lemma 1.9.12: f_n are measurable on a common domain E and $f_n \to f$ p.w. Assume E has finite measure. Then for any $\epsilon, \delta > 0$, there is a measurable subset $A \subseteq E$ and an index N such that $|f_n(x) - f(x)| < \epsilon$ for $x \in A$ when $n \geq N$, and $m(E - A) < \delta$.

Proof: First, by Proposition 1.9.9, f is measurable on E. Therefore, $|f_n - f|$ is measurable on E for every n.

Next, let $A_k = \{x \in E \mid |f_n(x) - f(x)| < \epsilon, \ \ n \geq k\}$. That is, A_k contains the points that $f_n(x)$ is sufficiently close to $f(x)$ when $n \geq k$. The points outside A_k need a larger k to guarantee convergence.

It is not difficult to see that A_k is measurable, since it is a countable intersection of measurable sets.

Now note that $E = \bigcup_{k=1}^{\infty} A_k$, because $f_n \to f$ p.w. on E. By the continuity of Lebesgue measure, we have

$$m(E) = m(\bigcup_{k=1}^{\infty} A_k) = \lim_{k \to \infty} m(A_k)$$

. $m(A_k)$ is an increasing sequence and has a limit. Therefore, there is a A_K such that $m(E) - m(A_K) < \delta$. Take $A = A_K$, and the K is the desired index. By the excision property, $m(E - A) = m(A) - E(E) < \delta$.

QED

Proposition 1.9.13 (Egoroff's Theorem): Assume E has finite measure. f_n are measurable on E and $f_n \to f$ p.w. on E. Then for any $\delta > 0$, there exists a closed subset $F \subseteq E$ such that $f_n \to f$ uniformly on F, and $m(E - F) < \delta$.

Proof: By Lemma 1.9.12, for any δ and n, there exists a measurable subset A_n and $k(n)$ such that $m(E - A_n) < \frac{\delta}{2^{n+1}}$, and $|f_j(x) - f(x)| < \frac{1}{n}$ for $x \in A_n, j \geq k(n)$.

Take $A = \bigcap_{n=1}^{\infty} A_n$. Then A is measurable. In addition,

$$m(E - A) = m(E - \bigcap_{n=1}^{\infty} A_n) = m(\bigcup_{n=1}^{\infty}(E - A_n)) \leq \sum_{n=1}^{\infty} \frac{\delta}{2^{n+1}} = \frac{\delta}{2}$$

We claim that $f_n \to f$ uniformly on A.

In fact, for any $\epsilon > 0$, choose a N such that $\frac{1}{N} < \epsilon$. Since $x \in A$, we have $x \in A_N$. There is an index $k(N)$ such that when $j \geq k(N)$, all $|f_j(x) - f(x)| < \frac{1}{N} < \epsilon$. Therefore, $f_n \to f$ uniformly on A.

Finally, by the inner approximation of Lebesgue measure, there is a closed set $F \subseteq A$ such that $m(A - F) < \frac{\delta}{2}$. Since $E - F = (E - A) \cup (A - F)$, we have $m(E - F) \leq m(E - A) + m(A - F) < \delta$.

QED

Lustin's Theorem

Lustin's Theorem states that for any measurable function f on E, we can find a closed subset $F \subseteq E$ such that: (1) f is continuous on F and (2) F is very close to E.

We first show it is true for simple functions, and then extend it to general cases.

Lemma 1.9.14: g is continuous on a closed set F. Then g can be extended to a continuous function on R .

Proof: Since F is closed, $E = R - F$ is open. Note that E can be written as a countable union of open intervals $\bigcup_{n=1}^{\infty}(a_n, b_n)$. Now define a new function h as follows:

$$h(x) = \begin{cases} g(x), & x \in F \\ f(a_n) + \frac{f(b_n) - f(a_n)}{b_n - a_n} \cdot f(b_n), & x \in (a_n, b_n) \end{cases}$$

Clearly, h is continuous on R.

QED

Proposition 1.9.15: f is a simple function E. Then for any $\delta > 0$, there is a closed subset $F \subseteq E$ such that: (1) $m(E - F) < \delta$, and (2) there is a continuous function g on R such that $f = g$ on F.

Proof: Let $a_1, \ldots, a_N$ be the values that f takes. Let $E_k = \{x \in E \mid f(x) = a_k\}, k = 1, 2, \ldots, N$. By inner approximation, for every E_k there is a closed subset $F_k \subseteq E_k$ such that $m(E_k - F_k) < \frac{\delta}{N}$. Let $F = \bigcup_{k=1}^{N} F_k$. Note that F is closed. Construct a function g on R for which $g(x) = a_k$ when $x \in E_k$. We claim g is continuous on F.

Observe that F_k are mutually disjoint. For any $x \in F$, $x \in F_k$ for some k. To prove that g is continuous on x, it suffices to show that there exists some d such that $(x-d, x+d) \cap F \subseteq F_k$. Suppose it is not the case. Then for every n, there exists $x_n \in (x - \frac{1}{n}, x + \frac{1}{n}) \cap F$ such that $x_n \notin F_k$. Note that $F' = F - F_k$ is also a closed set. The sequence $\{x_n\}$ satisfies that: (1) $x_n \in F'$ for every n, and (2) $x_n \to x$. As as result, it must be the case that $x \in F'$. This is a contradiction with the fact that $x \in F_k$.

So we have shown that g is continuous on F. Finally, by Lemma 1.9.14, we can extend g to R.

QED

Proposition 1.9.16 (Lustin's Theorem): f is a measurable function on E. Then for any $\delta > 0$, there is a closed subset $F \subseteq E$ such that: (1) $m(E-F) < \delta$, and (2) there is a continuous function g on R for which $f = g$ on F.

Proof: First, suppose $m(E) < \infty$.

Since f is measurable, by Simple Approximation Theorem, there is a sequence of simple functions f_n on E such that $f_n \to f$ p.w. on E. Then by Proposition 1.9.15, for every natural number k, there is a closed subset $F_k \subseteq E$ such that $m(E - F_k) < \frac{\delta}{2^{k+1}}$ and f_n is continuous on F_k for every n.

On the other hand, by Egoroff's Theorem, there is a closed subset $F_0 \subseteq E$ such that $m(E - F_0) < \frac{\delta}{2}$ and $f_n \to f$ uniformly on F_0.

Now, take $F = \bigcap_{k=0}^{\infty} F_k$. It is easy to see that $m(E-F) < \delta$. To show that f is continuous on F, note that

- f_n are continuous on F
- $f_n \to f$ uniformly on F

So consider any $x \in F$. For any $\epsilon > 0$, there is some d such that when $y \in (x-d, x+d) \cap F$, we have

$$(1) \;\; |f_n(y) - f_n(x)| < \frac{\epsilon}{3}, \;\; for\ every\ n$$

Since $f_n \to f$ uniformly on F, there is some N such that when $n > N$ we have

$$(2) \;\; |f(x) - f_n(x)| < \frac{\epsilon}{3}$$
$$(3) \;\; |f_n(y) - f(y)| < \frac{\epsilon}{3}$$

So combining (1)-(3), we choose a large n and when $y \in (x-d, x+d) \cap F$, we have

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| < \epsilon$$

. Therefore, f is continuous on F.

Next, consider the case $m(E) = \infty$. In this case, we can split E into a countable union of bounded sets. That is, let $E_n = E \cap [a_n, a_n + 1)$ where $n = 1, 2, \ldots$ and $a_n \in Z$. By the above arguments, there is a closed set $F_n \subseteq E_n$ for every n such that f is continuous on F_n and $m(E_n - F_n) < \frac{\delta}{2^n}$.

Now, take $F = \bigcup_{n=1}^{\infty} F_n$. It is easy to see that $m(E - F) < \delta$, and f is continuous on F. It remains to show that F is a closed set. To this end, consider a convergent sequence $x_k \to x$. Since $F \subseteq E$, $x \in E$. Then $x \in E_n$ for some n. Then for sufficiently large K, when $k \geq K$ we must have $\{x_k\}_{k \geq K}$ in $F_{n-1} \cup F_n$, which is a closed set. Therefore, their limit x must be in $F_{n-1} \cup F$. As a result, $x \in F$, which implies that F is closed.

QED

Excecise

Q1: f is a function defined on a measurable set E. Show that f is measurable if and only if for any Borel set B, $f^{-1}(B)$ is measurable.

Solution: We show that the sets B with the property that $f^{-1}(B)$ is Borel form a $\sigma-$algebra.

First, if $f^{-1}(B)$ is measurable, then $f^{-1}(R - B) = R - f^{-1}(B)$ is also measurable.

Next, if $f^{-1}(B_1)$ and $f^{-1}(B_2)$ are both measurable, then $f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$ is also measurable.

Finally, it is easy to see that both $f^{-1}(\emptyset)$ and $f^{-1}(R)$ are measurable.

Therefore, the sets with the property that $f^{-1}(B)$ is measurableform a σ-algebra. Let A be the $\sigma-$algebra.

Now, if f is measurable, by Proposition 1.9.2, for any open set O, $f^{-1}(O)$ is measurable. Then any open set $O \in A$. As a result, the Borel $\sigma-$algebra $\mathcal{B} \subseteq A$. So for any Borel set B, since $B \in A$, we have $f^{-1}(B)$ is measurable.

Conversely, if $f^{-1}(B)$ is measurable for any Borel set, then $f^{-1}(O)$ is measurable for any open set O. By Proposition 1.9.2, f is measurable.

QED

Q2. (Borel Measurability) A function f is said to be **Borel measurable** provided that it is defined on a Borel set E, and for any number c, the set $\{x \in E \mid f(x) < c\}$ is a Borel set. Show that

- (a) The following four statements are equivalent:
  - (i) For any real number c, the set $\{x \in E \mid f(x) < c\}$ is Borel
  - (ii) For any real number c, the set $\{x \in E \mid f(x) \leq c\}$ is Borel
  - (iii) For any real number c, the set $\{x \in E \mid f(x) > c\}$ is Borel
  - (iv) For any real number c, the set $\{x \in E \mid f(x) \geq c\}$ is Borel
- (b) If f, g are Borel measurable, then
  - (i) For any real number α, β, $\alpha \cdot f + \beta \cdot g$ is Borel measurable
  - (ii) $f \cdot g$ is Borel measurable
- (c) Every Borel measurable function is Lebesgue measurable.
- (d) If f is a Borel measurable function, and B is a Borel set, then $f^{-1}(B)$ is a Borel set.
- (e) If f, g are Borel measurable functions, then $f \circ g$ is also Borel measurable
- (f) If f is Borel measurable, g is Lebesgue measurable, then $f \circ g$ is Lebesgue measurable

Solution: (a) Note that E is a Borel set. For convenience, let $D_1(c), D_2(c), D_3(c), D_4(c)$ be the four sets of (i)-(iv), respectively.

For $(i) \Rightarrow (ii)$, note that $D_2(c) = \bigcap_{n=1}^{\infty} D_1(c + \frac{1}{n})$. Since $D_1(c + \frac{1}{n})$ is Borel for every n, $D_2(c)$ is Borel. The remaining parts are proved similarly.

(b) The proof is similar to the one of Proposition 1.9.6.

(c) Let f be a Borel measurable function on a Borel set E. Since every Borel set is Lebesgue measurable, E is a Lebesgue measurable set. Next, $f^{-1}(B)$ is Borel for any Borel set B. So $f^{-1}(B)$ is also Lebesgue measurable. By the solution of (d), f is Lebesgue measurable.

(d) Note that a Borel set can be constructed by applications of complement and countable unions on open sets. Since f is Borel measurable, by (a), $f^{-1}(O)$ is Borel for any open set O. Now consider a Borel B. It is constructed by applications of complementation and countable unions on open sets. Each of these pre-images are Borel. As a result, the pre-image $f^{-1}(B)$ is also Borel.

(e) For any Borel set B, since f, g are Borel measurable, by (d) we know that $(f \circ g)^{-1}(B)$ is a Borel set. Then the property (a) is easily satisfied, and $f \circ g$ is Borel measurable.

(f) For any Borel set B, since f is Borel measurable, $f^{-1}(B)$ is Borel. Since g is Lebesgue measurable, by Q1 we have $g^{-1}(f^{-1}(B))$ is Lebesgue measurable. So $(f \circ g)^{-1}(B)$ is Lebesgue measurable. Again, by Q1 we know that $f \circ g$ is Lebesgue measurable.

QED

Q3. Let $\{f_n\}$ be a sequence of measurable functions on E. Define E_0 to be the set that f_n converges. Is E_0 measurable?

Solution: For $x \in E_0$, since $f_n(x)$ converges, $f_n(x)$ is a Cauchy sequence. Thus, the set

$$E_0 = \bigcap_{1 \le n \le \infty} \bigcup_{1 \le m \le \infty} [\bigcap_{j,k \ge m} \{x \in E \mid \ |f_j(x) - f_k(x)| < \frac{1}{n}\}]$$

So E_0 is measurable.

QED

Q4. Let f be a bounded measurable function on E. Show that there are sequences of simple functions on E, $\{\phi_n\}$ and $\{\psi_n\}$ such that ϕ_n are increasing, ψ_n are decreasing and both ϕ_n and ψ_n converge to f uniformly on E.

Solution: By Simple Approximation Lemma, we construct the sequence of simple functions as follows. Since f is bounded, there exists some M such that $-M \le |f| \le M$. Take $n = 1$, and split $[-M, M]$ into intervals of $[a_k, a_k + 1)$, each of whose length is 1. So there exists simple functions ϕ_1 and ψ_1 for which $\phi_1 \le f \le \psi$ and $\psi_1 - \phi \le 1$. Next, take $n = 2$. And for each $[a_k, a_k + 1)$, split it into two intervals $[a_k, a_k + \frac{1}{2})$ and $[a_k + \frac{1}{2}, a_k + 1)$. Again there exist simple functions ϕ_2, ψ_2, corresponding to the new splitting of intervals and satisfies $\phi_2 \le f \le \psi_2$ and $\psi_2 - \phi_2 \le \frac{1}{2}$. Continuing this process, we construct the sequences of simple functions ϕ_n and ψ_n. Clearly, both ϕ_n and ψ_n converge to f uniformly on E. To show that ϕ_n is increasing, consider $x \in E$. If $\phi_n(x) \le f(x) \le \psi_n(x)$, then we split the interval$[\phi_n(x), \psi_n(x))$ into two intervals $[\phi_n(x), \phi_n(x) + \frac{1}{2})$ and $[\phi_n(x) + \frac{1}{2}, \psi_n)$. By the construction of ϕ_{n+1}, we know that if $f(x)$ is in the first

interval, $\phi_{n+1}(x) = \phi_n(x)$. If $f(x)$ is in the second interval, then $\phi_{n+1}(x) = \phi_n(x) + \frac{1}{2}$. Either way, we have $\phi_{n+1}(x) \geq \phi_n(x)$. Therefore, ϕ_n is increasing. Similarly, we can show that ψ_n is decreasing.

QED

Q5. A real-valued function is called *semi-simple* provided it takes only a countable number of values. Let f be any measurable function on E. Show that there is a sequence of semi-simple functions $\{f_n\}$ on E such that $f_n \to f$ uniformly on E.

Solution: If f is bounded, by Simple Approximation Lemma, we are done. Now suppose f is unbounded. The trick is to split $(-\infty, \infty)$ into a countable union of intervals $[-n, n)$ where $n \in Z$, and then construct functions

$$f_n(x) = \begin{cases} f(x), & x \in f^{-1}([-n, n)) \\ 0, & o.w. \end{cases}$$

It is easy to show that $f_n(x)$ is a bounded measurable function on E. Each n corresponds a sequence of uniformly convergent simple functions $\phi_{n,k}(x)$. It is straightforward to combine these functions to construct the sequence of semi-simple functions as desired.

QED

Q6. Let f be a measurable function that is finite a.e. on E and $m(E) < \infty$. For each $\epsilon > 0$, show that there is a measurable set $F \subseteq E$ such that $m(E - F) < \epsilon$, and f is bounded on F.

Solution: Applying Lustin's Theorem, there exists a closed subset $F \subseteq E$ such that $m(E - F) < \infty$ and f is continuous on F. Therefore, f is bounded on F.

QED

Q6. For a sequence of measurable functions $\{f_n\}$ on a common domain E, show that the following functions are all measurable:

- $\inf\{f_n\}$
- $\sup\{f_n\}$
- $\liminf\{f_n\}$
- $\limsup\{f_n\}$

Solution: (a) The set

$$\{x \in E \mid \inf\{f_n\}(x) \geq c\} = \bigcap_{1 \leq n \leq \infty} \{x \in E \mid f_n(x) \geq c\}$$

, and thus is measurable.

(b) Similar to (a).

(c) The set

$$\{x \in E \mid \liminf\{f_n\}(x) \geq c\} = \bigcup_{1 \leq k \leq \infty} \bigcap_{j \geq k} \{x \in E \mid f_j(x) \geq c\}$$

, and thus is measurable.

(d) Similar to (c)

QED

Q7. (Dini's Theorem) Let $\{f_n\}$ be an increasing sequence of continuous functions on $[a, b]$. $f_n \to f$ p.w. on $[a, b]$ and f is also a continuous function. Then $f_n \to f$ uniformly on $[a, b]$.

Proof: For any $\epsilon > 0$, let $E_n = \{x \in [a, b] \mid f(x) - f_n(x) < \epsilon\}$. We show that E_n is relatively open in $[a, b]$. That is, there is some open set A_n such that $E_n = A_n \cap [a, b]$. To this end, consider any $x \in E$. Let $\epsilon_0 = f(x) - f_n(x)$. Since f_n and f are continuous on $[a, b]$, there is some δ such that when $y \in (x - \delta, x + \delta) \cap [a, b]$, we have

$$|f(y) - f_n(y)| = |f(y) - f(x)| + |f(x) - f_n(x)| + |f_n(x) - f_n(y)| < \frac{\epsilon - \epsilon_0}{4} + \epsilon_0 + \frac{\epsilon - \epsilon_0}{4} < \epsilon$$

Therefore, $y \in E_n$.

Since f_n are increasing, so $E_n \subseteq E_{n+1}$, The corresponding open sets $A_n \subseteq A_{n+1}$. So $A_1, \ldots, A_n, \ldots$ is an open cover of $[a, b]$. Furthermore, since $f_n \to f$ p.w. on $[a, b]$, we have $\bigcup_n E_n = [a, b]$. By Heine-Borel Theorem, there is a finite number of E_n that covers $[a, b]$. Thus, there exists a E_N such that $E_N = [a, b]$. As a result, the point-wise convergence is also uniform on $[a, b]$.

QED